Deepa Parekh

FPGA Design Engineer

RESUME_1581

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Total Experience: 3 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 3 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: SPI, Verilog, UART, AXI4, FPGA, Xilinx Vivado, VHDL, Python, Intel Quartus, Hardware Simulation, Timing Closure, DSP

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

FPGA Design Engineer | LuminaForge Technologies | Jul 2023 - Present

- Reduced FPGA resource utilization by 22% through arithmetic sharing
- Verified 307 protocol corner cases with repeatable simulation suites
- Improved maximum operating frequency by 21% after timing-path optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

PROJECTS

Real-Time Signal Filter | DSP, Python, Verilog, Timing Closure

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Camera Data Bridge | VHDL, Xilinx Vivado, Hardware Simulation, FPGA

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Silverlake School of Engineering, 2023
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Digital Signal Processing Basics (2025)